

Rayyan Salman

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Education

Georgia Institute of Technology | Atlanta, GA, USA
Bachelor of Science in Mechanical Engineering, GPA 3.59/4.00

Expected Graduation: 05/28

Experience

LATTICEPT, Roswell, GA

January - July 2026

Mechanical Engineer Intern

- Developed a FANUC CRX-10iA/L automation platform integrating ROS 2 and KAREL with Stream Motion (J519) for Python-based trajectory generation and 8 ms UDP motion control.
- Implemented 3-point coordinate calibration, path interpolation, and Cartesian motion workflows for autonomous path execution, robotic writing, and future welding applications.
- Diagnosed FANUC motion faults and tuned trajectory velocity, acceleration, jerk, and command timing to achieve stable high-frequency robot motion.
- Integrated a SERVO-ROBOT FUJI-CAM machine-vision system with robot hardware and developed Python tools to export, visualize, and analyze weld-profile scan data for geometry comparison and weld-quality assessment.
- Created 3D CAD models and simulation-ready geometry for GPU-native lattice Boltzmann CFD workflows, supporting verification, post-processing, and engineering analysis.

VIP: High Altitude Balloon Research, Georgia Institute of Technology

August 2025 - December 2025

August 2026 - Present

Undergraduate Researcher

- Developed Python scripts to parse, clean, and structure 1,800+ timestamped atmospheric sensor records, including 4-channel electric-field, temperature, pressure, and UV data.
- Developed CAN-FD payload communications with MCP2518/RP2040 hardware using SocketCAN and PyCyphal.
- Designed and integrated sensor housings, PCB mounts, and flight-ready payload assemblies while supporting mechanical, electrical, and sensor-system integration and testing.

Skills

CAD/Design: SolidWorks (CSWA), Engineering Drawings, GD&T, Fusion 360, CATIA

Robotics & Automation: FANUC CRX, Stream Motion (J519), ROS 2, KAREL, Motion Planning, Machine Vision

Programming: Python, MATLAB, Java, Arduino

Embedded & Data: CAN-FD, SocketCAN, PyCyphal, RP2040, pandas, NumPy, Matplotlib

Simulation: CFD, FEA, Lattice Boltzmann Methods, Pre/Post-Processing

Fabrication / Tools: 3D Printing, TIG Welding, Laser Cutting, CNC Machining, Excel

Engineering Projects

Search and Rescue Ant Robot | Team Lead, Mechanical Design

August - December 2025

- Led mechanical design of a multi-legged robotic platform for search-and-rescue operations in confined environments.
- Created SolidWorks assemblies and detailed engineering drawings with GD&T for linkages, joints, and sensor housings.
- Performed design trade studies and motion evaluations to assess mobility, stability, and manufacturability.

Adaptive Power Wheelchair Modification

2023

- Redesigned seating, controls, structural supports, and mounting points on a Power Wheels car for a child with cerebral palsy, improving accessibility and safety.

Involvement

FIRST Robotics Competition (FRC), Alpharetta, GA

2022 - 2024

Mechanical Design Team

- Designed drivetrain and intake assemblies in SolidWorks and translated designs into fabricated hardware through CNC machining and 3D printing.